

Extreme Heat and Urban Mobility: Evidence from Mobile Network Data in Spain

Andrew Renninger*

Carmen Cabrera†

January 2026

Abstract

Cities rely on dense, repeated movement—to workplaces, shops, and social venues—to generate face-to-face contact, match quality, and local consumption. Rising extreme heat may therefore operate as a recurring negative shock to urban interaction intensity. We study how heat affects daily mobility using district-by-day origin–destination flows for ~ 13 million mobile devices in Spain (about 27% of the population), provided by the Spanish Ministry for Transport and Sustainable Mobility, combined with high-resolution thermal-comfort measures from ERA5-HEAT (Universal Thermal Climate Index, UTCI). Using Poisson models with district and date fixed effects, we find that extreme heat substantially reduces discretionary mobility: when $UTCI > 40^\circ\text{C}$, trips to infrequently visited destinations fall by about 10%, and by up to 20% during the hottest afternoon hours. In contrast, work/study travel is comparatively resilient on average, but declines for older adults and higher-income groups. Lower-income groups reduce discretionary trips but show little reduction in work/study travel, consistent with constraints on avoidance. We also document that heat reshapes urban interaction patterns: the income diversity of visitors to destinations falls on very hot days and gravity estimates indicate that mobility losses are concentrated in dense urban cores. The results imply that extreme heat can depress the very movements that support agglomeration and urban consumption, while shifting exposure risk toward groups with limited flexibility.

JEL codes: Q54, R41, R23, J22.

Keywords: extreme heat; urban mobility; amenity demand; agglomeration; inequality; adaptation.

1 Introduction

Extreme heat is no longer a rare “tail” event in Southern Europe. It is increasingly a predictable feature of the summer, with consequences for health and mortality [Masselot et al., 2023, Vicedo-Cabrera et al., 2021], labor productivity [Somanathan et al., 2021, Colacito et al., 2019], and economic growth [Dell et al., 2014]. For urban economists, however, a different margin is equally important: extreme heat may systematically change the intensity and spatial pattern of daily movement inside and between cities. That movement underpins the micro-foundations of agglomeration—meetings, shopping, services, networking, and incidental contact [Glaeser, 2011, Arzaghi and Henderson, 2008, Atkin et al., 2022]. If heat curbs mobility, it can reduce urban consumption and

*Center for Advanced Spatial Analysis, University College London. Email: andrew.renninger.12@ucl.ac.uk.

†Geographic Data Science Lab, University of Liverpool. Email: c.cabrera@liverpool.ac.uk.

weaken the mechanisms that generate productivity and innovation, even when measured economic output is not immediately observed.

This paper asks three empirical questions. First, how does extreme heat change the *level* of daily mobility, and does the response differ between routine and discretionary travel? Second, does heat affect mobility *unequally* across demographic and socioeconomic groups, potentially shifting exposure and economic burdens? Third, does heat reshape the *structure* of mobility networks—who visits which places—in ways that matter for urban interaction and segregation?

We answer these questions using a large-scale mobility product from the Spanish Ministry for Transport and Sustainable Mobility [Ministry of Transport, Mobility and Urban Agenda of Spain, MITMA, 2024], which aggregates passive and active mobile network events into daily origin–destination flows between 3,599 districts. The data cover roughly 13 million devices (about 27% of Spain’s population) and are pre-stratified by trip purpose (work/study; frequent destinations; infrequent destinations), time of day, trip distance, age band, gender, and household-income class. We merge these flows with the Universal Thermal Climate Index (UTCI) from ERA5-HEAT [Di Napoli et al., 2021], a thermal-comfort metric that incorporates temperature, humidity, wind, and solar radiation.

Our empirical design exploits short-run variation in heat within districts over time, while controlling for district fixed effects and date fixed effects. The date effects absorb nationwide shocks and seasonal patterns (including holidays and the August vacation season), so identification comes from differential exposure to heat across districts on the same date. We estimate Poisson models (appropriate for count outcomes and common in gravity and mobility settings) and present coefficients as semi-elasticities in flows.

The results are economically meaningful. When $UTCI > 40^{\circ}\text{C}$, discretionary mobility falls sharply: trips to infrequently visited destinations decline by about 10% overall, and by up to 20% during hot afternoons when thermal stress peaks. In contrast, work/study travel is stable on average, consistent with work obligations and institutional constraints. Heterogeneity reveals a distributional pattern: older adults reduce mobility the most and are the only group with clear declines in work/study travel, while lower-income groups show little reduction in work/study mobility even on extremely hot days—suggesting limited ability to avoid exposure by staying home or working remotely. We find no meaningful differences by gender after conditioning on trip purpose. Beyond trip counts, we show that heat reduces the income diversity of visitors to destinations and that network losses are concentrated in dense urban cores, consistent with heat acting as a negative shock to downtown activity and interaction.

The paper contributes to several literatures. First, it complements work documenting economic impacts of temperature on productivity and growth [Dell et al., 2014, Colacito et al., 2019] by focusing on the mobility margin that links climate to urban interactions. Second, it adds large-scale evidence on behavioral adaptation to heat, beyond small-sample time-use or travel-survey studies [Böcker et al., 2013, Batur et al., 2024]. Third, it speaks to distributional and inequality concerns in climate adaptation: flexibility in mobility (and work) is not evenly distributed, and heat may shift burdens toward groups with fewer avoidance options [Cong et al., 2022, Barreca et al., 2016].

The remainder of the paper proceeds as follows. Section 2 situates the work in urban and climate economics. Section 3 describes the mobility and climate data. Section 4 lays out the empirical strategy and identification assumptions. Section 5 presents the main results and heterogeneity. Section 6 interprets the findings through an urban-economics lens. Section 7 summarizes robustness and validation checks. The Appendix contains detailed data construction, additional results, placebo tests, gravity-model estimates, and an illustrative (no-adaptation) climate-projection exercise.

2 Background and related literature

A core theme in climate economics is that temperature affects welfare and output through multiple channels, including mortality, morbidity, and productivity [Masselot et al., 2023, Somanathan et al., 2021, Dell et al., 2014]. A parallel theme in urban economics is that the value of cities comes from interaction—matching, learning, and consumption in dense places [Glaeser, 2011, Arzaghi and Henderson, 2008]. Extreme heat plausibly connects these literatures by changing *where* and *whether* people travel on a given day.

Existing evidence on heat and mobility often relies on travel surveys, small geographic contexts, or specific modes (e.g., cycling) [Böcker et al., 2013, 2019, Basu et al., 2024]. Time-use studies show substitution away from outdoor activities on hot days [Graff Zivin and Neidell, 2014, Batur et al., 2024], but they rarely observe citywide network structure or heterogeneity by socioeconomic status at scale. Recent work demonstrates the value of passively collected mobility traces for studying behavioral responses to shocks [Hatchett et al., 2021, González-Leonardo et al., 2024, Ly et al., 2023], including climate-related disruptions.

Our focus on Spain is motivated by three features relevant for urban and climate adaptation. First, Southern Europe is projected to face increased heat extremes [Yin et al., 2019, Wu et al., 2024, Lorenzo et al., 2021]. Second, population aging and chronic disease prevalence increase vulnerability [Xu et al., 2014, Zanobetti et al., 2012]. Third, relatively low air-conditioning penetration compared to the U.S. implies that behavioral adaptation and place-based cooling policies may matter more [International Energy Agency, 2018]. In this setting, extreme heat can shift mobility and thereby affect urban consumption and interaction patterns.

3 Data and measurement

3.1 Mobility flows from mobile network data

We use daily origin–destination mobility data provided by the Spanish Ministry for Transport and Sustainable Mobility [Ministry of Transport, Mobility and Urban Agenda of Spain, MITMA, 2024]. The Ministry constructs aggregated OD matrices by combining anonymized mobile network events (active events such as calls/texts and passive network “pings”) with demographic balancing and validation against administrative and survey benchmarks (see Appendix A for details and documentation references). The key advantages for our purposes are: (i) national coverage, (ii) high temporal regularity (daily), (iii) consistent spatial units (districts), and (iv) stratification by trip purpose and demographics.

The unit of geography is the district (*distrito*), with 3,599 districts covering mainland Spain and the Balearic Islands. We observe aggregated flows *within* districts and *between* districts; we do not observe individual trajectories.

Trip purposes are provided as three mutually exclusive categories:

- **Work/study:** trips to destinations inferred to be work or school locations.
- **Frequent:** destinations visited more than once within a two-week window.
- **Infrequent:** destinations visited only once within the same two-week window.

The work/study label may capture some routine non-employment activities for certain groups if mobility patterns are recurrent but not employment-related; we treat it as “routine/obligatory” travel in interpretation and verify that results are robust across alternative samples and checks (Appendix D).

The analysis sample covers the warm season (May–September) in 2022 and 2023. We focus on this window because our research question is about extreme heat and because many mobility patterns are structurally different in winter. We exclude international travel to keep the analysis on domestic mobility.

3.2 Heat exposure: thermal comfort (UTCI)

We measure heat exposure using ERA5-HEAT reanalysis data [Di Napoli et al., 2021]. Our primary measure is the Universal Thermal Climate Index (UTCI), which combines temperature, humidity, wind speed, and solar radiation to capture thermal stress. For each district and day, we compute mean UTCI at 16:00 local time (a standard time close to peak daily heat stress in Spain during summer). Appendix A provides construction details.

We study both (i) a binary extreme-heat indicator ($\text{UTCI} > 40^\circ\text{C}$) and (ii) binned UTCI exposures in 5°C increments (25–30, 30–35, 35–40, 40–45, 45–50), with 25–30 as the reference bin in the main heterogeneity specifications. The binned approach is useful for economists because it makes nonlinearity and threshold-like behavior explicit without requiring functional-form assumptions.

3.3 Outcome variables

Our main outcomes are daily trip counts by district and purpose. We also study (i) time-of-day and distance slices of trips, (ii) mixing patterns across districts using origin-population and origin-income deciles, and (iii) bilateral flows in a gravity-model framework (used for both interpretation and identification diagnostics).

Table 1 reports descriptive statistics by demographic strata (full detail in the Appendix). Figures 1–5 preview core patterns.

4 Empirical strategy

4.1 Baseline fixed-effects Poisson model

Let T_{ita} denote the number of trips terminating in district i on date t for activity (purpose) category $a \in \{\text{work/study, frequent, infrequent}\}$. We model T_{ita} as a count outcome and estimate a fixed-effects Poisson (PPML) specification:

$$\mathbb{E}[T_{ita} \mid X_{it}] = \exp\left(\alpha_i + \gamma_t + \sum_b \beta_{ab} \mathbf{1}\{\text{UTCI}_{it} \in b\}\right), \quad (1)$$

where α_i are district fixed effects and γ_t are date fixed effects. The indicators $\mathbf{1}\{\text{UTCI}_{it} \in b\}$ correspond to UTCI bins (or, in simpler specifications, a single indicator for $\text{UTCI}_{it} > 40^\circ\text{C}$). In binned models, the omitted category is $\text{UTCI} \in [25^\circ\text{C}, 30^\circ\text{C})$, so coefficients β_{ab} are semi-elasticities relative to that baseline (i.e., $\beta \approx -0.10$ implies roughly a 10% decline).

We estimate (1) separately for within-district and between-district flows where noted. Standard errors are clustered at the province level in district-level models to allow for spatial correlation in unobservables.

4.2 Identification and interpretation

With district fixed effects, identification comes from within-district deviations in heat over time. With date fixed effects, identification further comes from differences in heat exposure *across districts on the same date*. This design flexibly absorbs national shocks (weekends, holidays, macro

Table 1: Descriptive statistics by age, income, and gender (May–September 2022–2023)

Stratum	Activity	Trips (billions)	Share (%)	Obs. (millions)	Districts
<i>Panel A: Age group</i>					
0–25	Work/Study	0.51	5.0	1.02	3,599
	Frequent	1.49	14.6	1.08	3,599
	Infrequent	0.48	4.7	1.09	3,599
25–45	Work/Study	0.63	6.2	1.06	3,599
	Frequent	1.69	16.6	1.09	3,599
	Infrequent	0.55	5.4	1.10	3,599
45–65	Work/Study	0.67	6.6	1.08	3,599
	Frequent	1.95	19.2	1.10	3,599
	Infrequent	0.61	6.0	1.10	3,599
65–100	Work/Study	0.21	2.1	0.90	3,594
	Frequent	1.07	10.5	1.08	3,599
	Infrequent	0.30	2.9	1.08	3,599
<i>Panel B: Household income (thousands of euros)</i>					
< 10	Work/Study	0.53	3.9	0.83	3,586
	Frequent	1.55	11.4	0.90	3,599
	Infrequent	0.47	3.4	1.00	3,599
10–15	Work/Study	1.76	12.9	1.10	3,599
	Frequent	5.28	38.7	1.10	3,599
	Infrequent	1.63	11.9	1.10	3,599
> 15	Work/Study	0.48	3.5	0.77	3,500
	Frequent	1.46	10.7	0.90	3,599
	Infrequent	0.50	3.6	1.02	3,599
<i>Panel C: Gender</i>					
Female	Work/Study	0.83	9.9	1.03	3,599
	Frequent	2.70	32.2	1.09	3,599
	Infrequent	0.81	9.7	1.09	3,599
Male	Work/Study	0.79	9.5	1.04	3,599
	Frequent	2.51	29.9	1.09	3,599
	Infrequent	0.74	8.8	1.10	3,599

Notes: Trips are summed over the sample period. “Obs.” denotes district–day cells for each stratum and activity type (in millions). Small differences in district counts reflect missing observations in some strata for Work/Study.

news) and national seasonality (including the August vacation period). Under the assumption that, conditional on (α_i, γ_t) , short-run variation in district-level UTCI is orthogonal to unobserved district-level shocks to mobility, β_{ab} can be interpreted as the causal effect of heat exposure on mobility for activity a .

Two threats are worth emphasizing.

- **Time-varying local confounders:** Some local events could correlate with heat (e.g., festivals that occur in hotter months). Date fixed effects eliminate confounding from events common across Spain; remaining bias would require events that are both district-specific and correlated with deviations in heat within district.
- **Spatial substitution (SUTVA concerns):** If heat in one district causes people to travel to another district to seek cooler conditions, the “treatment” could change the distribution of destinations. We test for such patterns in a gravity model that includes temperature gradients; we find no evidence that flows systematically shift toward cooler destinations during heat extremes (Appendix F).

4.3 Nonlinear responses and descriptive functional form

For visualization and to recover smooth nonlinear patterns, we also estimate generalized additive models (GAMs) that flexibly relate mobility to UTCI while controlling for day-of-year trends and day-of-week/holiday effects. We treat the GAM as descriptive evidence on functional form; the fixed-effects Poisson models are the primary basis for causal interpretation (Appendix B).

4.4 Mixing metrics and network structure

To capture how heat reshapes interaction patterns (not only levels), we construct two metrics at the destination-district-by-day level, separately for origin-income and origin-population deciles:

- **Richness:** number of distinct origin deciles that send any visitors to a destination on a day.
- **Evenness:** normalized Shannon entropy of the distribution of origin deciles among visitors.

We then estimate fixed-effects models analogous to (1), replacing trip counts with these mixing measures (Appendix B).

5 Results

5.1 Heat reduces discretionary travel much more than routine travel

Figure 1 summarizes the baseline pattern. Panel C shows two-way fixed-effects estimates using a binary heat indicator ($UTCI > 40^\circ\text{C}$): infrequent (discretionary) trips fall by roughly 10%, frequent trips fall by about 2–3%, and work/study travel shows no meaningful average change. Panel D shows a smooth functional form from the GAM: mobility increases modestly as conditions move from cool to warm (consistent with outdoor activity being more attractive), but beyond high thermal stress, mobility declines.

An important economic interpretation is that extreme heat disproportionately reduces *non-routine* movement—the trips most plausibly linked to consumption amenities, leisure, and spontaneous social interaction. Routine commuting is less elastic, consistent with institutional constraints and the fixed timing of many jobs and schooling.

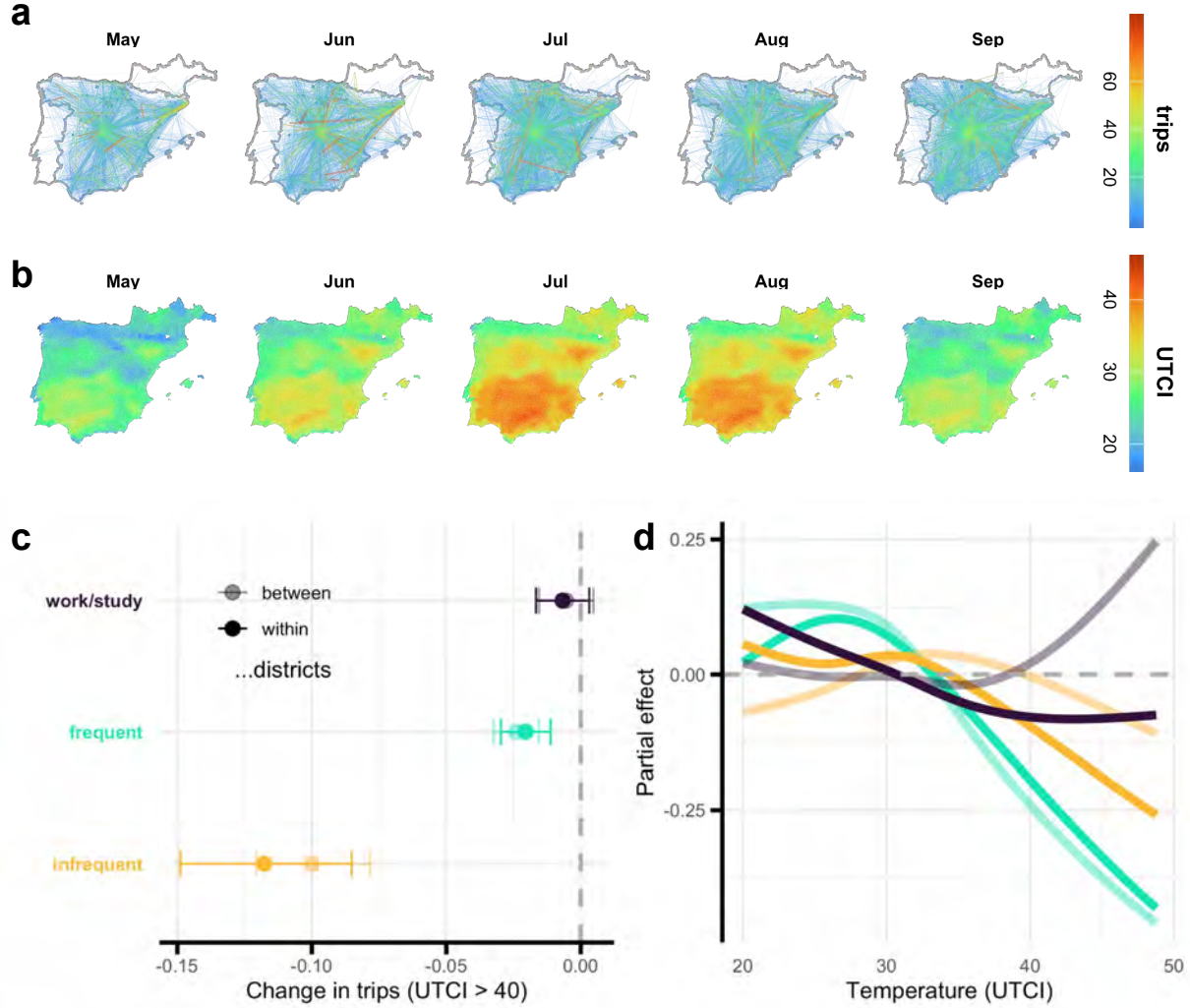


Figure 1: **Extreme heat and mobility levels.** **A** Monthly mobility networks in Spain (2022–2023 summers), demonstrating national coverage (network mass concentrated in major metros). **B** Mean UTCI by month, with peaks in July–August. **C** Two-way fixed-effects Poisson estimates of the effect of UTCI > 40°C on trip counts by activity type (semi-elasticities; 95% CI). Discretionary (infrequent) mobility falls most. **D** GAM estimates treating UTCI as continuous show an “inverted-U” relationship for discretionary activities (activity rises with warmth until very high thermal stress, then declines); work/study is comparatively flat.

5.2 Heterogeneity: age and income gradients, little gender difference

Figure 2 reports binned-temperature estimates by demographic groups. The age gradient is sharp: older adults reduce mobility the most, and heat-related declines show up even in work/study travel for the oldest group. This is consistent with higher health risk and lower thermal tolerance at older ages [Xu et al., 2014, Oudin Åström et al., 2011, Zanobetti et al., 2012].

Income heterogeneity suggests constraints on avoidance. We observe a clear reduction in discretionary (infrequent) travel for all groups as heat increases. But lower-income groups show little reduction in work/study travel even at high UTCI, while higher-income groups reduce mobility across activity types, including work/study. A plausible interpretation is that higher-income workers have more flexibility (including remote work) and/or can better substitute away from travel, whereas lower-income workers face higher costs of missing work. We treat this as evidence of behavioral constraints rather than direct evidence on remote work, which we do not observe.

Finally, after conditioning on trip purpose, we find no meaningful gender differences in heat responses.

5.3 The largest effects occur in hot afternoons and on short trips

Heat is not constant within the day. Figure 3 shows that the strongest mobility reductions occur during the afternoon, when thermal stress peaks. Infrequent travel can fall by up to 20% in hot afternoons. We do not see strong evidence of complete substitution to mornings or evenings: mobility still declines during those periods on very hot days, suggesting a net reduction rather than only rescheduling.

We also find stronger declines for short trips (e.g., < 2 km), consistent with heat discouraging walking/cycling and other active travel [Böcker et al., 2013]. Longer trips are more resilient, potentially because they are more likely to involve motorized (and possibly air-conditioned) transport.

5.4 Heat reduces socioeconomic mixing and shifts activity away from the urban core

Mobility is also a network that links neighborhoods and social groups. Figure 4 shows descriptive “mixing matrices” and fixed-effects estimates for diversity metrics. Under normal thermal conditions, trips are biased toward high-population destinations (an “urban pull”) and toward higher-income destinations (a “wealth pull”). As temperatures rise, these biases attenuate: movements become relatively more concentrated within mid-density and mid-income areas, and the income diversity of visitors to a destination falls on very hot days.

From an urban-economics perspective, this matters because interaction across neighborhoods and social strata is a key mechanism for learning, matching, and consumption variety. Heat-induced segmentation can therefore reduce the cross-cutting contacts that help cities generate spillovers.

Finally, Figure 5 uses a gravity framework to formalize how heat interacts with distance and the urban hierarchy. The results imply that, as temperatures rise from 35 °C to 40 °C, mobility losses are larger for shorter-distance links and for flows involving highly populated destinations. Visualizing predicted network changes suggests that urban cores lose the most activity on hot days.

6 Implications for urban economics and policy

The empirical patterns can be read as evidence that extreme heat operates like a recurring negative shock to urban interaction intensity, with three implications.

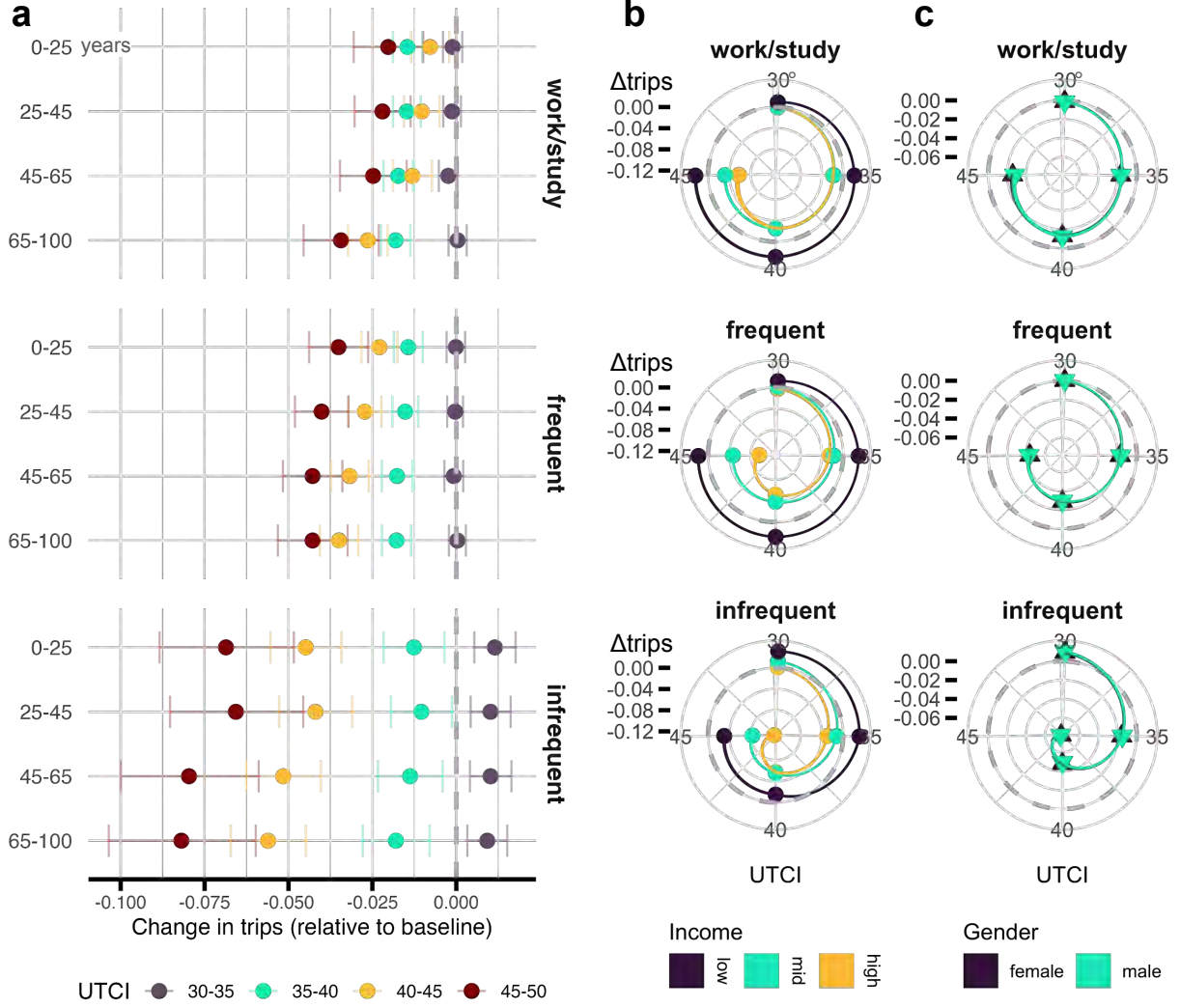


Figure 2: **Heterogeneity by demographics.** Effects are relative to the omitted UTCI $\in [25^{\circ}\text{C}, 30^{\circ}\text{C})$ bin and are semi-elasticities (log points). **A** Age: heat reduces mobility for all ages, with larger declines for older adults; the oldest show the clearest reductions in work/study travel. **B** Income: lower-income groups curb discretionary mobility but show little reduction in work/study travel at high UTCI; higher-income groups reduce mobility across activities, consistent with greater avoidance options. **C** Gender: responses are statistically indistinguishable once trip purpose is held fixed.

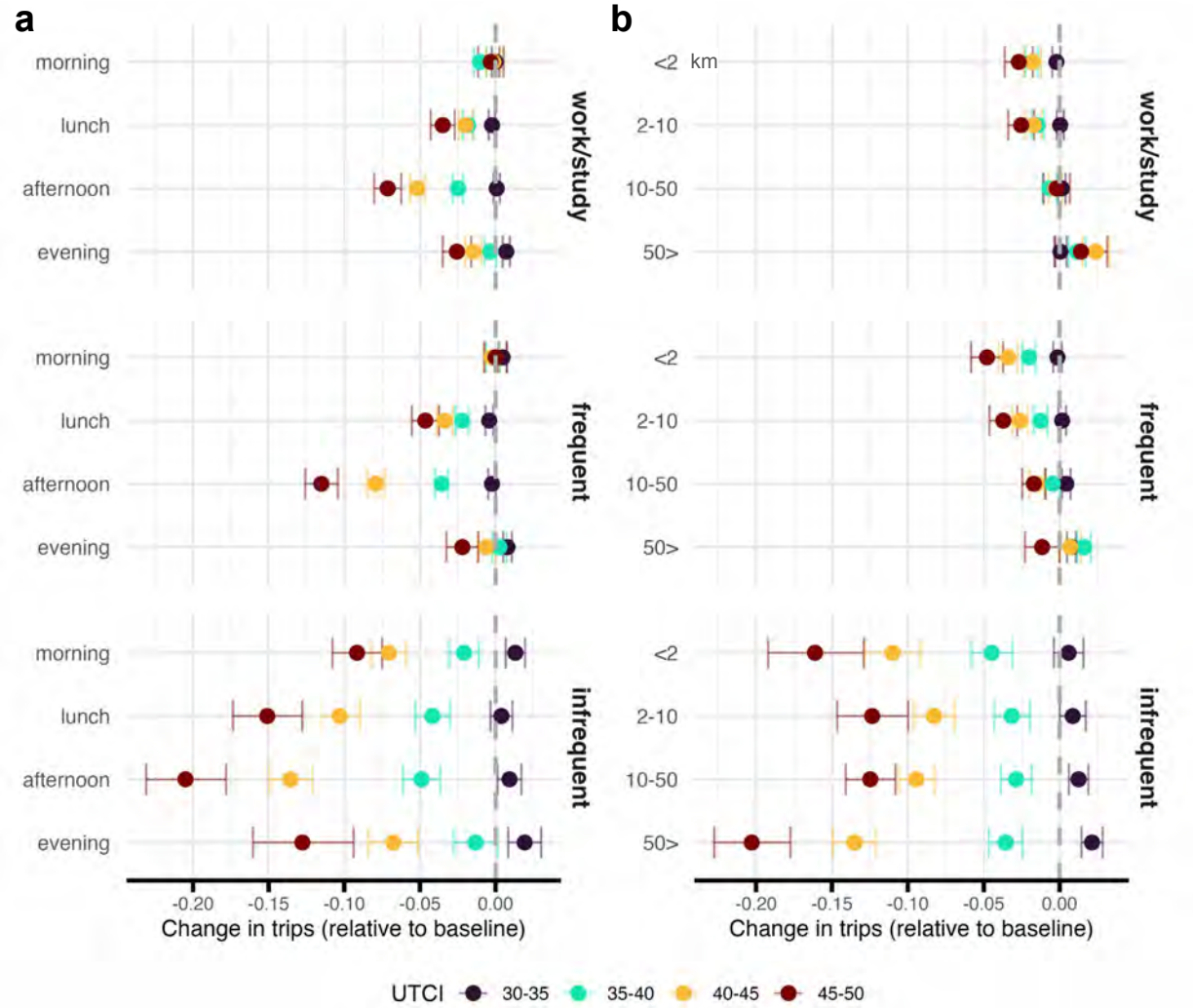


Figure 3: **Time of day and trip distance.** Effects are relative to $UTCI \in [25^{\circ}\text{C}, 30^{\circ}\text{C})$ and are semi-elasticities. **A** The largest declines occur in the afternoon when thermal stress peaks; there is limited evidence of full intertemporal substitution to cooler hours. **B** Short trips decline more than long trips, consistent with heat deterring active travel.

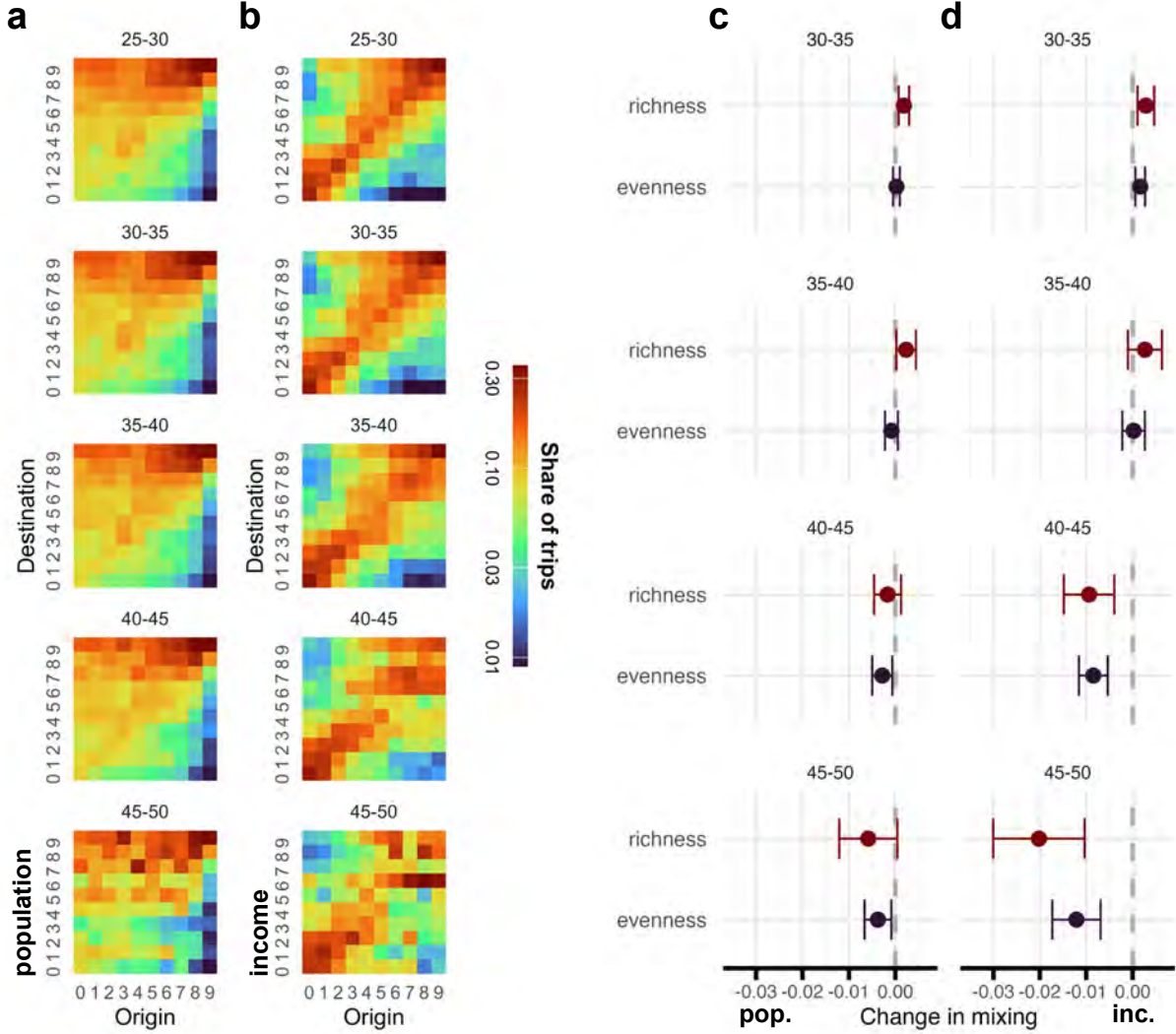


Figure 4: **Mixing and network structure under heat.** **A** Flows between districts ordered by population decile: under moderate conditions, mobility is oriented toward high-population destinations; this pattern weakens at high UTCI. **B** Flows ordered by district income decile: under moderate conditions, trips skew toward higher-income destinations; this also weakens with heat. **C–D** Fixed-effects estimates using diversity metrics (richness and evenness) show that income diversity of visitors declines on very hot days (semi-elasticities relative to $UTCI \in [25^\circ\text{C}, 30^\circ\text{C})$).

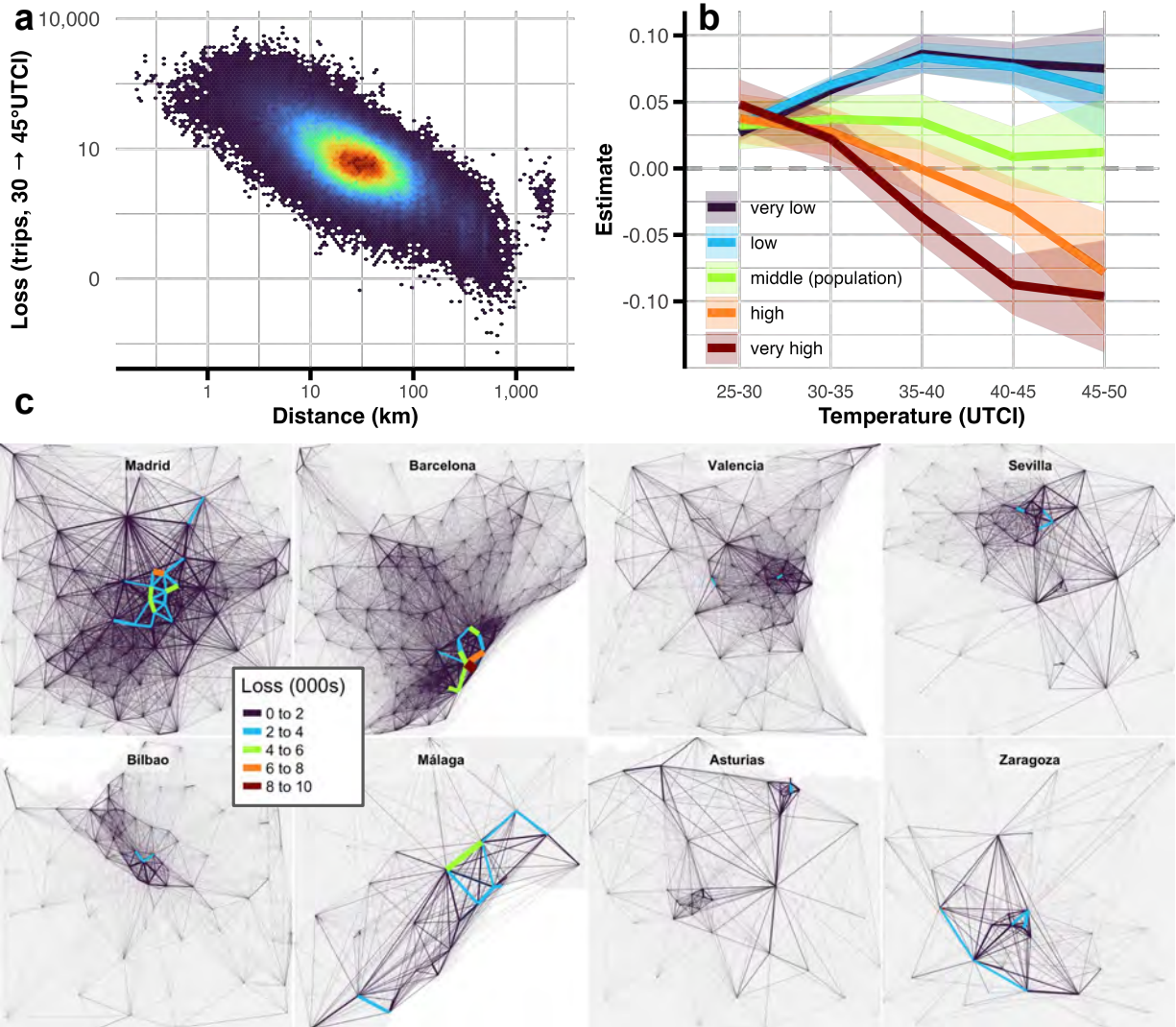


Figure 5: **Urban structure and gravity-model implications.** **A** Estimated losses by distance when UTCI rises from 35 °C to 40 °C: short links decline more. **B** Interacting temperature with population quantiles shows larger declines for flows involving high-population (core) destinations. **C** Network visualization of predicted changes suggests that mobility losses concentrate in the urban core rather than the periphery.

1. Heat is a “demand shock” to discretionary urban activity. The largest elasticities occur for infrequent trips and in hot afternoons. These are precisely the movements most plausibly linked to consumption amenities (retail, hospitality, culture) and spontaneous social contact. Even if people continue to work, fewer ancillary trips can reduce footfall and local demand. This channel can help explain why heat waves may affect urban service-sector activity even when measured commuting changes are modest.

2. Heat reallocates exposure and constraint across groups. Older adults reduce mobility more strongly, consistent with health risk. Lower-income individuals appear less able to reduce work/study travel, suggesting that heat may shift exposure toward those with fewer avoidance options. This is a distributional problem: behavioral adaptation is not free and is mediated by labor-market constraints and the ability to work remotely or reschedule activities.

3. Heat reshapes where interaction happens. The evidence that diversity of visitors declines and that losses concentrate in cores is consistent with a weakening of core-periphery interaction on hot days. If extreme heat becomes more frequent and adaptation remains incomplete, this could erode some benefits of urban density—not because density is intrinsically bad, but because the “contact technology” of cities becomes harder to use when ambient conditions are hostile.

These findings underscore the potential complementarity between *building-level* adaptation (air conditioning, remote work) and *place-based* adaptation (shading, greening, reflective materials, cool corridors) [Jay et al., 2021]. Building-level cooling can protect health but may also reduce street-level interaction if it encourages retreat indoors. Place-based cooling has the potential to preserve outdoor mobility and the interactions that make cities productive and socially vibrant.

7 Robustness and validation

The Appendix reports a set of checks designed to address common econometric concerns.

- **Seasonality and the August vacation period:** We re-estimate core models excluding August and find closely similar qualitative results (Appendix D, Tables A2–A4).
- **Placebo tests:** Shuffling UTCI across dates or across districts produces near-zero effects, supporting the interpretation that the main estimates reflect heat exposure rather than spurious time or space patterns (Appendix E, Figure A5).
- **City-size and year subsamples:** GAM patterns and core conclusions are stable when restricting to large cities versus smaller places and when estimating separately by year (Appendix D, Figure A4).
- **Spatial substitution:** Gravity models including temperature gradients show no meaningful evidence that, during heat waves, people systematically re-route trips toward cooler districts in a way that would overturn the fixed-effects interpretation (Appendix F, Table A5).

8 Conclusion

Extreme heat changes not only health and productivity but also the micro-behavioral patterns that make cities function. Using large-scale mobile-network mobility data in Spain combined with thermal-stress measures, we show that extreme heat substantially reduces discretionary urban

mobility, especially during hot afternoons, while routine work/study travel is comparatively resilient on average. The heterogeneity results indicate that older adults reduce mobility the most and that lower-income groups appear less able to reduce work/study travel, suggesting that adaptation constraints can shift exposure and burdens in unequal ways. We also document that heat reshapes mobility networks by reducing socioeconomic mixing and concentrating losses in dense urban cores.

For urban economics, the implication is that heat can operate as a repeated, predictable disruption to the interaction intensity that underlies agglomeration and urban consumption. A natural next step is to connect these mobility disruptions to outcomes such as retail sales, transit revenue, workplace attendance, or productivity measures, and to evaluate which place-based cooling interventions best preserve socially valuable movement under rising heat stress.

References

- Mohammad Arzaghi and J Vernon Henderson. Networking off madison avenue. *The Review of Economic Studies*, 75(4):1011–1038, 2008.
- David Atkin, M Keith Chen, and Anton Popov. The returns to face-to-face interactions: Knowledge spillovers in silicon valley. Technical report, National Bureau of Economic Research, 2022.
- Alan Barreca, Karen Clay, Olivier Deschenes, Michael Greenstone, and Joseph S Shapiro. Adapting to climate change: The remarkable decline in the us temperature-mortality relationship over the twentieth century. *Journal of Political Economy*, 124(1):105–159, 2016.
- Rounaq Basu, Nicola Colaninno, Aziz Alhassan, and Andres Sevtsuk. Hot and bothered: Exploring the effect of heat on pedestrian route choice behavior and accessibility. *Cities*, 155:105435, 2024.
- Irfan Batur, Victor O Alhassan, Mikhail V Chester, Steven E Polzin, Cynthia Chen, Chandra R Bhat, and Ram M Pendyala. Understanding how extreme heat impacts human activity-mobility and time use patterns. *Transportation Research Part D: Transport and Environment*, 136:104431, 2024.
- Lars Böcker, Martin Dijst, and Jan Prillwitz. Impact of everyday weather on individual daily travel behaviours in perspective: a literature review. *Transport reviews*, 33(1):71–91, 2013.
- Lars Böcker, Tanu Priya Uteng, Chengxi Liu, and Martin Dijst. Weather and daily mobility in international perspective: A cross-comparison of dutch, norwegian and swedish city regions. *Transportation research part D: transport and environment*, 77:491–505, 2019.
- Riccardo Colacito, Bridget Hoffmann, and Toan Phan. Temperature and growth: A panel analysis of the united states. *Journal of Money, Credit and Banking*, 51(2-3):313–368, 2019.
- Shuchen Cong, Destenie Nock, Yueming Lucy Qiu, and Bo Xing. Unveiling hidden energy poverty using the energy equity gap. *Nature communications*, 13(1):2456, 2022.
- Melissa Dell, Benjamin F Jones, and Benjamin A Olken. What do we learn from the weather? the new climate-economy literature. *Journal of Economic literature*, 52(3):740–798, 2014.
- Claudia Di Napoli, Christopher Barnard, Christel Prudhomme, Hannah L Cloke, and Florian Pappenberger. Era5-heat: a global gridded historical dataset of human thermal comfort indices from climate reanalysis. *Geoscience data journal*, 8(1):2–10, 2021.

- Edward Glaeser. Triumph of the city, 2011.
- Miguel González-Leonardo, Carmen Cabrera-Arnau, and Francisco Rowe. Harnessing digital data to strengthen disaster response in valencia’s 2024 floods. *Population Europe*, 2024.
- Joshua Graff Zivin and Matthew Neidell. Temperature and the allocation of time: Implications for climate change. *Journal of Labor Economics*, 32(1):1–26, 2014.
- Benjamin J Hatchett, Tarik Benmarhnia, Kristen Guirguis, Kristin VanderMolen, Alexander Gershunov, Heather Kerwin, Andrey Khlystov, Kathryn M Lambrecht, and Vera Samburova. Mobility data to aid assessment of human responses to extreme environmental conditions. *The Lancet Planetary Health*, 5(10):e665–e667, 2021.
- Garth Heutel, Nolan H Miller, and David Molitor. Adaptation and the mortality effects of temperature across us climate regions. *Review of Economics and Statistics*, 103(4):740–753, 2021.
- International Energy Agency. The future of cooling. Technical report, International Energy Agency, 2018. URL <https://www.iea.org/reports/the-future-of-cooling>. Accessed on November 29, 2024.
- Ollie Jay, Anthony Capon, Peter Berry, Carolyn Broderick, Richard de Dear, George Havenith, Yasushi Honda, R Sari Kovats, Wei Ma, Arunima Malik, et al. Reducing the health effects of hot weather and heat extremes: from personal cooling strategies to green cities. *The Lancet*, 398(10301):709–724, 2021.
- Nieves Lorenzo, Alejandro Díaz-Poso, and Dominic Royé. Heatwave intensity on the iberian peninsula: Future climate projections. *Atmospheric Research*, 258:105655, 2021.
- Robert E Lucas Jr. Econometric policy evaluation: A critique. In *Carnegie-Rochester conference series on public policy*, volume 1, pages 19–46. North-Holland, 1976.
- Anthony Ly, Frances V Davenport, and Noah S Diffenbaugh. Exploring the influence of summer temperature on human mobility during the covid-19 pandemic in the san francisco bay area. *GeoHealth*, 7:e2022GH000772, 2023.
- Pierre Masselot, Malcolm Mistry, Jacopo Vanoli, Rochelle Schneider, Tamara Iungman, David Garcia-Leon, Juan-Carlos Ciscar, Luc Feyen, Hans Orru, Aleš Urban, et al. Excess mortality attributed to heat and cold: a health impact assessment study in 854 cities in europe. *The Lancet Planetary Health*, 7(4):e271–e281, 2023.
- Ministry of Transport, Mobility and Urban Agenda of Spain, MITMA. Open Data Movilidad 2020-2021. <https://www.transportes.gob.es/ministerio/proyectos-singulares/estudios-de-movilidad-con-big-data/opendata-movilidad>, 2024.
- Daniel Oudin Åström, Forsberg Bertil, and Joacim Rocklöv. Heat wave impact on morbidity and mortality in the elderly population: A review of recent studies. *Maturitas*, 69(2):99–105, 2011.
- Eswaran Somanathan, Rohini Somanathan, Anant Sudarshan, and Meenu Tewari. The impact of temperature on productivity and labor supply: Evidence from indian manufacturing. *Journal of Political Economy*, 129(6):1797–1827, 2021.
- Bridget Thrasher, Weile Wang, Andrew Michaelis, Forrest Melton, Tsengdar Lee, and Ramakrishna Nemani. Nasa global daily downscaled projections, cmip6. *Scientific data*, 9(1):262, 2022.

- Ana Maria Vicedo-Cabrera, N Scovronick, Francesco Sera, Dominic Royé, Rochelle Schneider, Aurelio Tobias, Christofer Astrom, Y Guo, Y Honda, DM Hondula, et al. The burden of heat-related mortality attributable to recent human-induced climate change. *Nature climate change*, 11(6):492–500, 2021.
- Yao Wu, Bo Wen, Antonio Gasparrini, Ben Armstrong, Francesco Sera, Eric Lavigne, Shanshan Li, Yuming Guo, Ala Overcenco, Aleš Urban, et al. Temperature frequency and mortality: Assessing adaptation to local temperature. *Environment international*, 187:108691, 2024.
- Zhiwei Xu, Ruth A Etzel, Hong Su, Cunrui Huang, Yuming Guo, and Shilu Tong. The impact of heat waves on children’s health: a systematic review. *International Journal of Biometeorology*, 58:239–247, 2014.
- Qian Yin, Jinfeng Wang, Zhoupeng Ren, Jie Li, and Yuming Guo. Mapping the increased minimum mortality temperatures in the context of global climate change. *Nature communications*, 10(1):4640, 2019.
- Antonella Zanobetti, Marie S O’neill, Carina J Gronlund, and Joel D Schwartz. Summer temperature variability and long-term survival among elderly people with chronic disease. *Proceedings of the National Academy of Sciences*, 109(17):6608–6613, 2012.

A Data construction and additional descriptives

A.1 Mobility data: Ministry OD product

We use daily origin–destination data provided by the Spanish Ministry for Transport [[Ministry of Transport, Mobility and Urban Agenda of Spain, MITMA, 2024](#)], which integrates anonymized mobile network records with demographic balancing and validation against official statistics. The dataset captures trips over 500 m within Spain and provides aggregated matrices by spatial unit and day. The Ministry assigns trip purpose into (i) work/study, (ii) frequent, and (iii) infrequent destinations based on recurrent mobility patterns over rolling two-week windows.

Compared to app-based GPS data, mobile network data typically have broader population coverage and different selection properties because they do not require a specific application to be installed. The Ministry also applies balancing procedures so that the distribution of devices matches official population distributions by age, income, and region, reducing representativeness concerns.

We collect the data for May–September 2022 and 2023. National COVID restrictions were largely lifted by early 2022; we focus on 2022–2023 to avoid pandemic-era mobility artifacts.

A.2 Heat exposure: UTCI from ERA5-HEAT

We link mobility to a thermal comfort index from ERA5-HEAT reanalysis data [[Di Napoli et al., 2021](#)]. UTCI combines temperature, wind, radiation, and humidity to measure thermal stress. For each district and day, we compute mean UTCI at 16:00 using zonal statistics.

A.3 Additional descriptive patterns

Figures [A1–A3](#) visualize time-series patterns and correlations that motivate the fixed-effects specification (strong weekday/weekend structure; strong seasonality and an August vacation shift; scaling of work/study flows with population).

B Empirical details and additional methods

B.1 Two-way fixed effects (district and date) Poisson models

Our main approach models trip counts as Poisson:

$$T_{ita} \sim \text{Poisson}(\mu_{ita}), \quad \log(\mu_{ita}) = \alpha_i + \gamma_t + \sum_b \beta_{ab} \mathbf{1}\{\text{UTCI}_{it} \in b\}.$$

We implement (i) a binary $\text{UTCI} > 40^\circ\text{C}$ specification for summary effects and (ii) binned UTCI specifications for heterogeneity and nonlinearities.

B.2 Placebo tests

We test for spurious associations by permuting heat exposure in two ways: (i) shuffle UTCI across districts within a date (breaking spatial structure while keeping national time patterns), (ii) shuffle UTCI across dates within a district (breaking time structure while keeping district averages). Estimated effects collapse toward zero under both permutations (Figure [A5](#)).

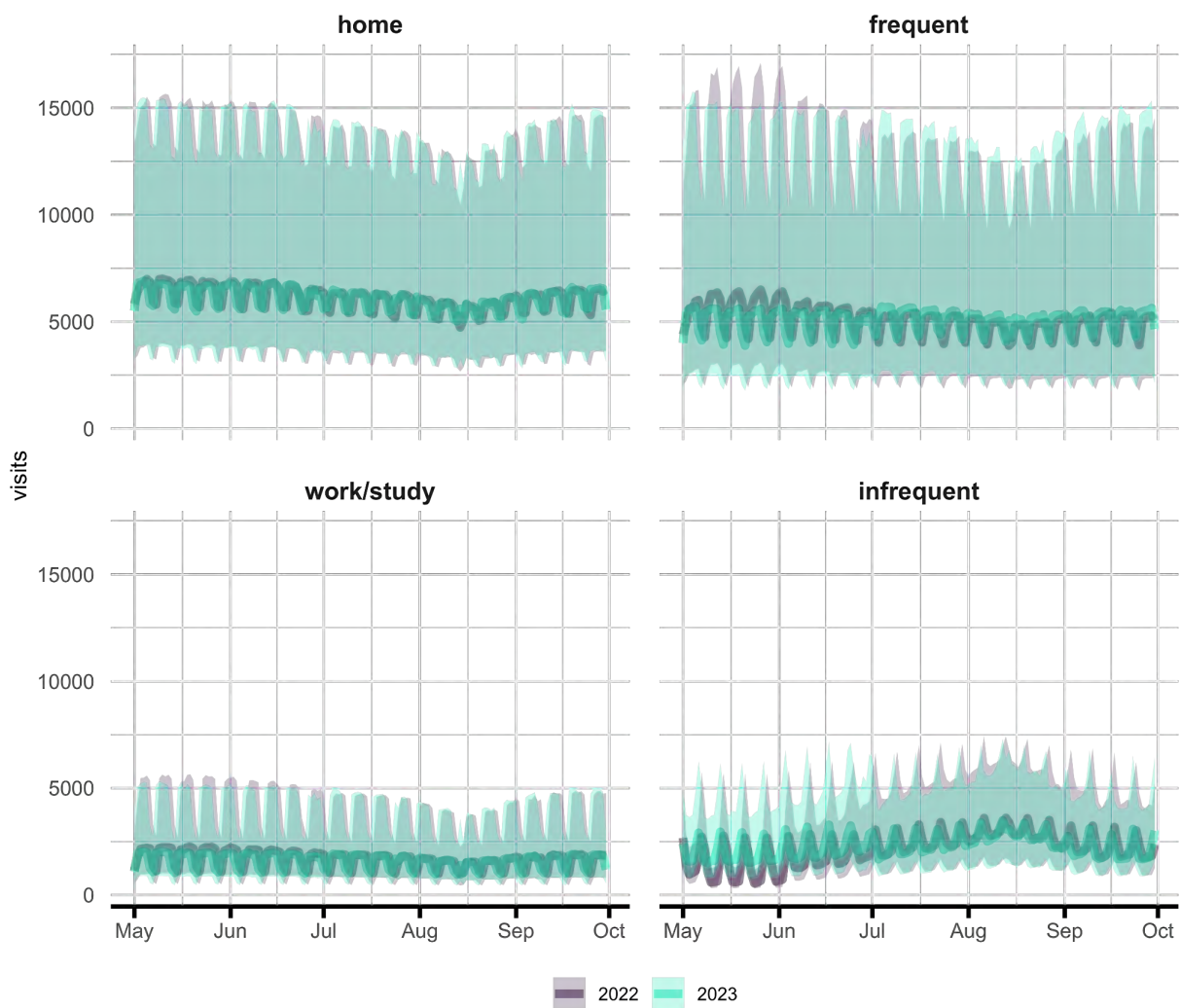


Figure A1: **Trends in mobility by activity.** Daily time series over the two summers with interquartile ranges. Work/study and frequent activity are more common on weekdays; infrequent activity rises on weekends and in August. These patterns motivate flexible time controls (date fixed effects) in the main econometric design.

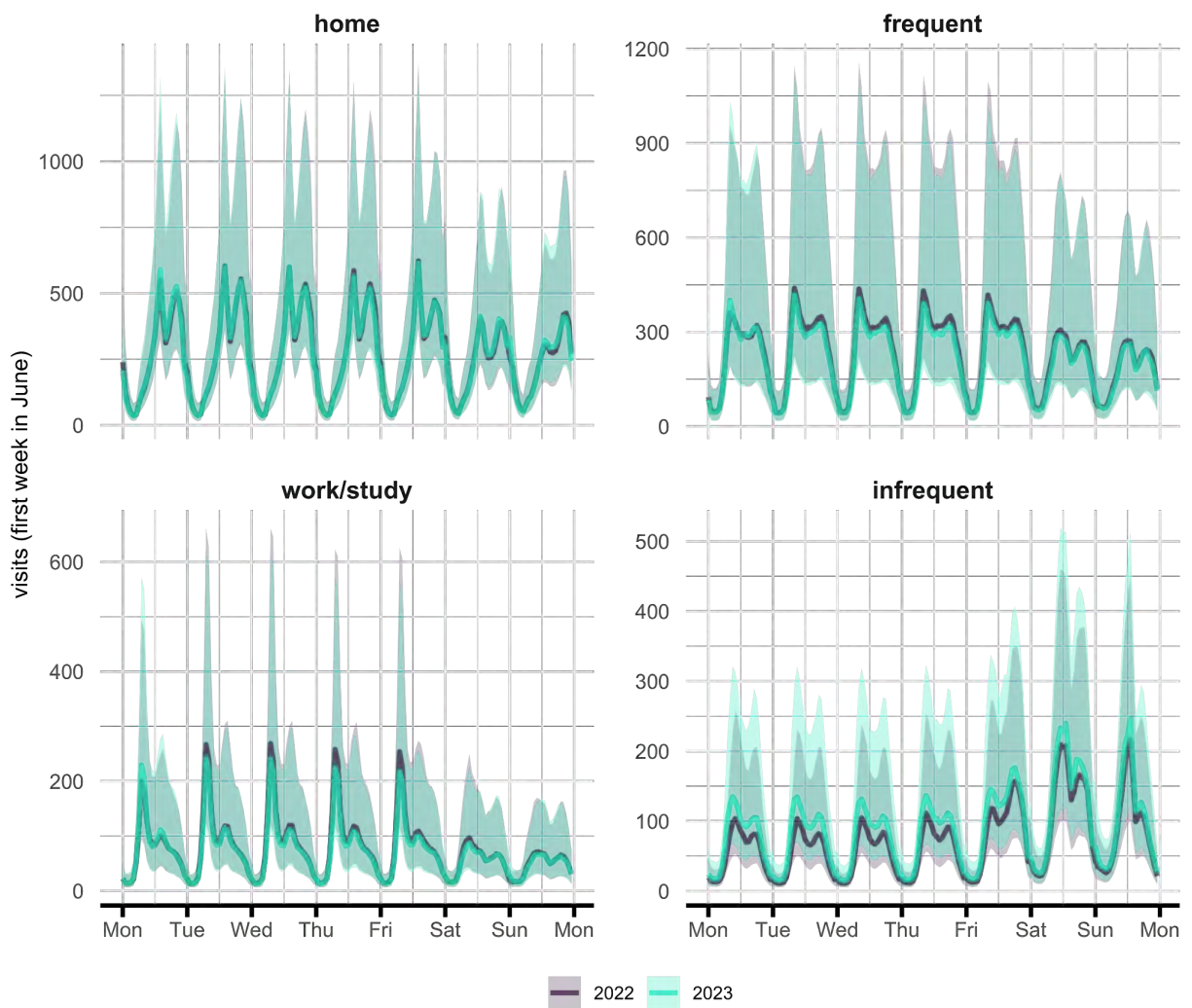


Figure A2: **Weekly patterns in mobility.** Example week in early June. Work/study shows a strong weekday rhythm, consistent with commuting.

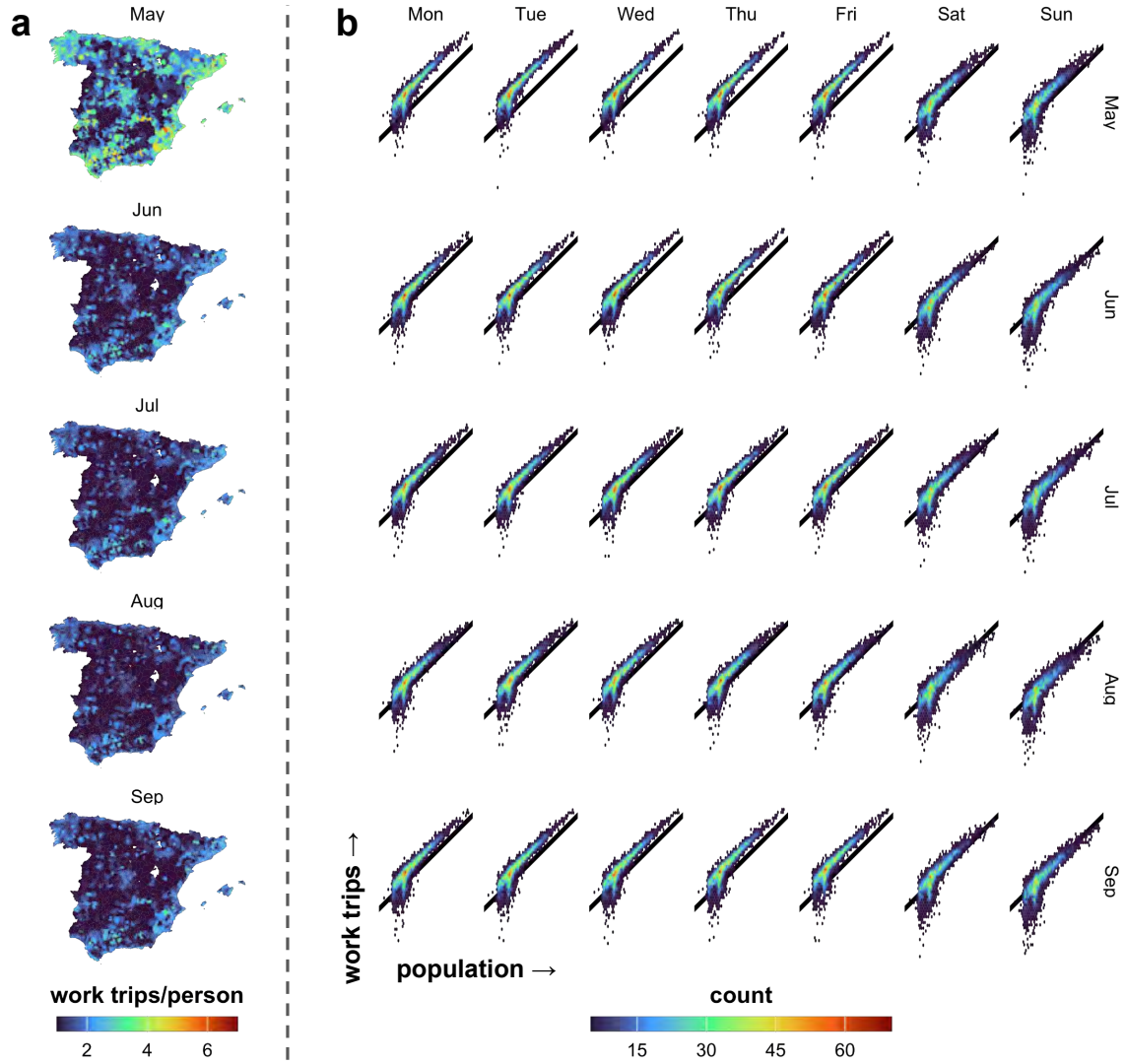


Figure A3: **Mobility and population.** **A** Work/study trips per person by district and month. **B** Correlation between work/study trips and population; the relationship differs in May (school term effects and classification nuances). We address these patterns with fixed effects and robustness checks.

B.3 Generalized additive model (GAM)

To visualize nonlinear responses, we estimate a GAM with smooths:

$$\log(\mu_{ita}) = f_1(\text{UTCI}_{it}) \times a + f_2(\text{day-of-year}_t) + \text{DoW}_t + \text{holiday}_t + \text{controls}.$$

We treat the GAM as descriptive; causal interpretation relies on the fixed-effects models.

B.4 Mixing metrics

For each destination district i and day t , we compute:

- Richness R_{it} : number of distinct origin groups (income or population deciles) that send trips to i on day t .
- Evenness E_{it} : normalized Shannon entropy:

$$E_{it} = -\frac{\sum_{g=1}^{R_{it}} p_{git} \ln(p_{git})}{\ln(R_{it})},$$

where p_{git} is the share of visitors from group g .

We regress these measures on UTCI bins with district and date fixed effects, clustering by province.

C Main regression tables

Table A1: Baseline effect of extreme heat (UTCI > 40 °C) on mobility by trip purpose

Dependent variable: flows	Within district	Between districts
Work/Study	-0.0068 (0.0087)	-0.0054 (0.0079)
Frequent	-0.0207** (0.0086)	-0.0243*** (0.0069)
Infrequent	-0.125*** (0.0311)	-0.105*** (0.0232)
District FE	Yes	Yes
Date FE	Yes	Yes
Pseudo R^2	0.978	0.972

Notes: Poisson regressions with district and date fixed effects. Standard errors clustered by province in parentheses. Semi-elasticities in expected counts. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D Robustness: excluding August

Spain’s August vacation season is a potential concern because both mobility and heat are high. Date fixed effects already absorb nationwide August patterns, but we additionally re-estimate key binned models excluding August and show that conclusions are stable.

Table A2: Heat effects by age and trip purpose: robustness to excluding August

Activity	Sample	UTCI bin (°C)			
		30–35	35–40	40–45	45–50
Panel A: Age 0–25					
Frequent	All months	-0.0002**	-0.0145***	-0.0232***	-0.0358***
	Excl. Aug.	0.0001	-0.0123***	-0.0232***	-0.0435***
Infrequent	All months	0.0114***	-0.0130***	-0.0463***	-0.0707***
	Excl. Aug.	0.0098***	-0.0136***	-0.0495***	-0.0937***
Work/Study	All months	-0.0012***	-0.0148***	-0.0082***	-0.0207***
	Excl. Aug.	0.0004**	-0.0114***	-0.0092***	-0.0206***
Panel B: Age 25–45					
Frequent	All months	-0.0004***	-0.0154***	-0.0276***	-0.0410***
	Excl. Aug.	0.0006***	-0.0126***	-0.0297***	-0.0500***
Infrequent	All months	0.0101***	-0.0109***	-0.0434***	-0.0683***
	Excl. Aug.	0.0092***	-0.0112***	-0.0437***	-0.0846***
Work/Study	All months	-0.0013***	-0.0149***	-0.0104***	-0.0222***
	Excl. Aug.	0.0011***	-0.0110***	-0.0117***	-0.0195***
Panel C: Age 45–65					
Frequent	All months	-0.0008***	-0.0178***	-0.0322***	-0.0438***
	Excl. Aug.	-0.0001	-0.0131***	-0.0325***	-0.0537***
Infrequent	All months	0.0101***	-0.0142***	-0.0533***	-0.0831***
	Excl. Aug.	0.0094***	-0.0130***	-0.0543***	-0.1026***
Work/Study	All months	-0.0025***	-0.0175***	-0.0133***	-0.0251***
	Excl. Aug.	0.0002	-0.0122***	-0.0141***	-0.0279***
Panel D: Age 65–100					
Frequent	All months	0.0003***	-0.0180***	-0.0355***	-0.0438***
	Excl. Aug.	0.0011***	-0.0131***	-0.0328***	-0.0524***
Infrequent	All months	0.0088***	-0.0193***	-0.0591***	-0.0863***
	Excl. Aug.	0.0092***	-0.0162***	-0.0607***	-0.1067***
Work/Study	All months	0.0003	-0.0185***	-0.0265***	-0.0345***
	Excl. Aug.	0.0017***	-0.0136***	-0.0248***	-0.0441***

Notes: Coefficients from Poisson regressions of within-district flows on UTCI-bin indicators interacted with age and activity. District and date fixed effects included; standard errors clustered by province (not shown). Reference bin: 25–30°C. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A3: Heat effects by income and trip purpose: robustness to excluding August

Activity	Sample	UTCI bin (°C)			
		30–35	35–40	40–45	45–50
Panel A: Income < 10					
Frequent	All months	0.0124***	0.0300***	0.0295***	0.0202***
	Excl. Aug.	0.0134***	0.0352***	0.0379***	0.0335***
Infrequent	All months	0.0312***	0.0311***	-0.0139***	-0.0295***
	Excl. Aug.	0.0263***	0.0263***	-0.0236***	-0.0502***
Work/Study	All months	0.0096***	0.0220***	0.0305***	0.0260***
	Excl. Aug.	0.0095***	0.0279***	0.0390***	0.0418***
Panel B: Income 10–15					
Frequent	All months	0.0008***	-0.0177***	-0.0391***	-0.0480***
	Excl. Aug.	0.0003***	-0.0158***	-0.0385***	-0.0587***
Infrequent	All months	0.0122***	-0.0124***	-0.0573***	-0.0873***
	Excl. Aug.	0.0098***	-0.0155***	-0.0623***	-0.1127***
Work/Study	All months	-0.0001*	-0.0178***	-0.0251***	-0.0324***
	Excl. Aug.	0.0011***	-0.0132***	-0.0214***	-0.0323***
Panel C: Income > 15					
Frequent	All months	-0.0028***	-0.0255***	-0.0525***	-0.0997***
	Excl. Aug.	-0.0057***	-0.0264***	-0.0549***	-0.1131***
Infrequent	All months	0.0014***	-0.0296***	-0.0575***	-0.1326***
	Excl. Aug.	-0.0002	-0.0250***	-0.0453***	-0.1351***
Work/Study	All months	-0.0010***	-0.0188***	-0.0252***	-0.0590***
	Excl. Aug.	-0.0012***	-0.0204***	-0.0308***	-0.0700***

Notes: Same structure as Table A2. Reference bin: 25–30°C.

Table A4: Heat effects on mixing metrics: robustness to excluding August

Metric	Sample	UTCI bin (°C)			
		30–35	35–40	40–45	45–50
Panel A: Population-based mixing					
Evenness	All months	0.00019	-0.00059	-0.00201**	-0.00267**
	Excl. Aug.	0.00009	-0.00028	-0.00143**	-0.00209
Richness	All months	0.0173***	0.0218**	-0.0154	-0.0541*
	Excl. Aug.	0.0167***	0.0150	-0.0162	-0.0680
Panel B: Income-based mixing					
Evenness	All months	0.00019	-0.00091	-0.00237**	-0.00110
	Excl. Aug.	0.00008	-0.00029	-0.00085	0.00160
Richness	All months	0.0227***	0.0212	-0.0749***	-0.1613***
	Excl. Aug.	0.0194***	0.0201	-0.0943***	-0.1543**

Notes: Entries are coefficients from fixed-effects regressions of district-day mixing metrics on UTCI bins (reference: 25–30°C). District and date fixed effects included; standard errors clustered by province (not shown). Significance:

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

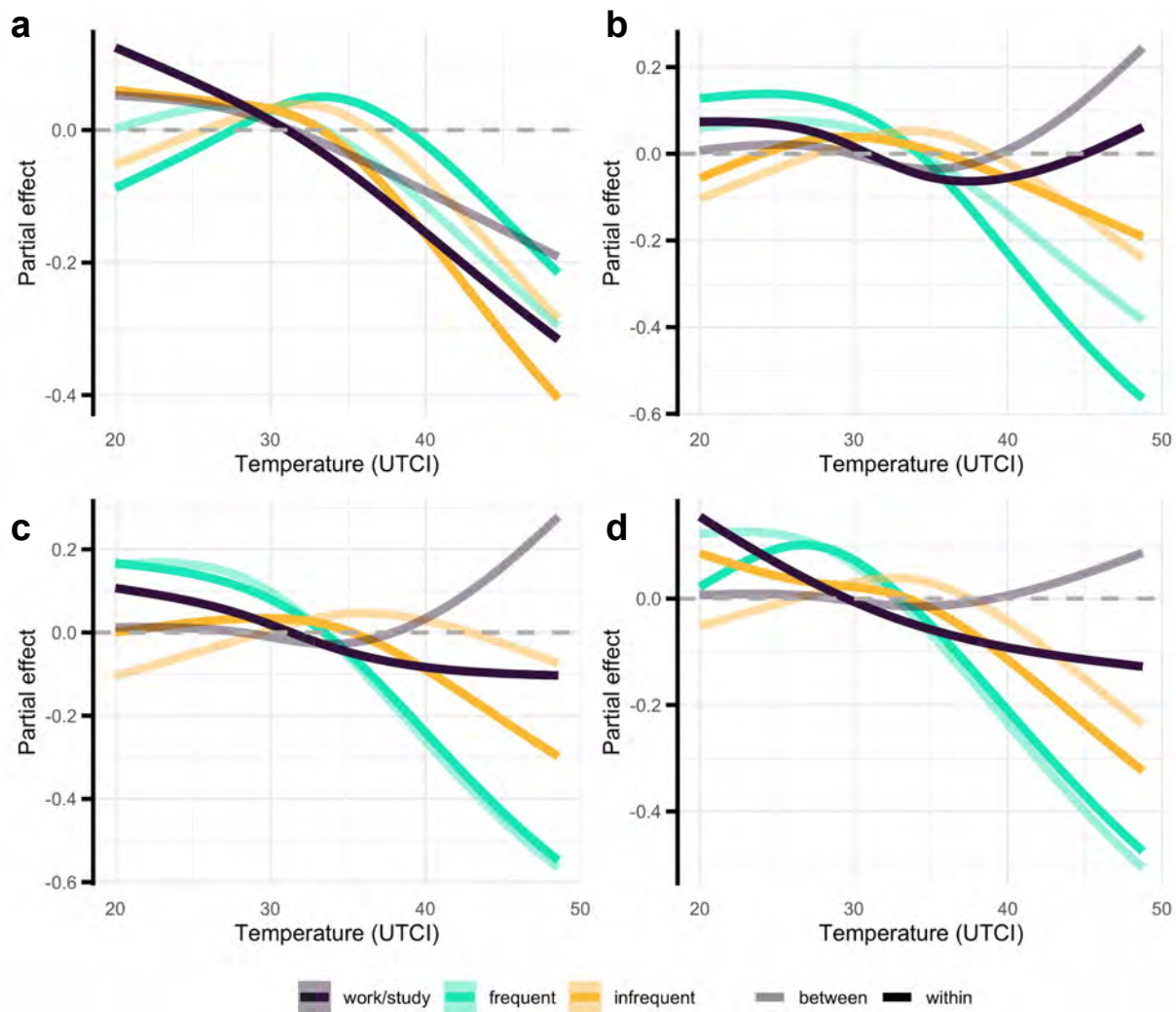


Figure A4: **Sensitivity checks.** GAM estimates for large cities versus smaller places and separately by year show broadly stable nonlinear patterns.

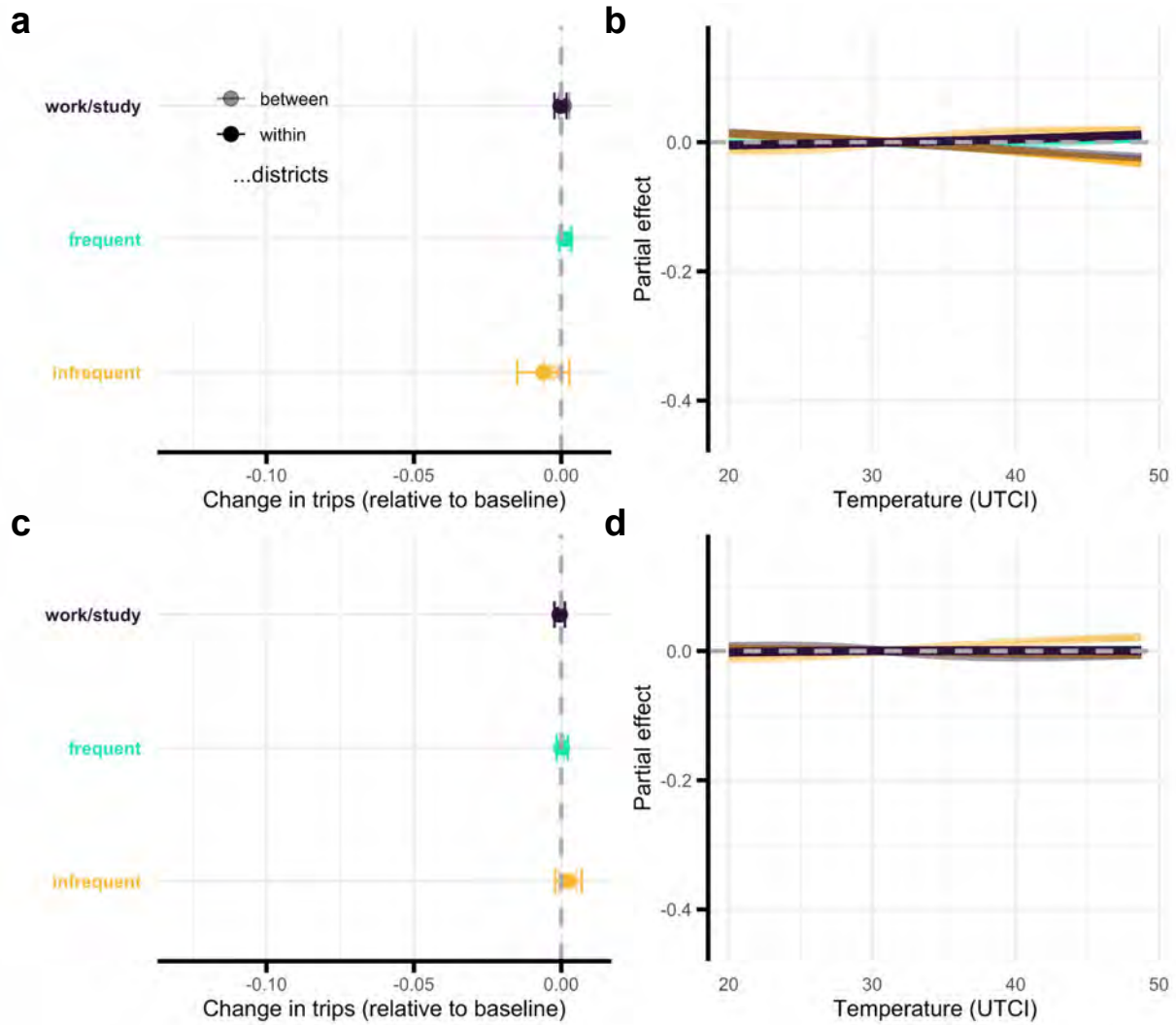


Figure A5: **Placebo tests.** We permute UTCI across districts within a date and across dates within a district. Estimated effects shrink substantially (often by an order of magnitude) relative to the main estimates, supporting the interpretation that the primary results reflect heat exposure rather than spurious patterns.

E Placebo tests

F Gravity models and spatial-substitution diagnostics

We use gravity models to (i) test for substitution toward cooler destinations that could threaten fixed-effects interpretation and (ii) quantify how heat modifies distance decay and core-periphery link strength.

Table A5: Gravity models: temperature bins and temperature-gradient diagnostic

Dependent variable: Trips	Base	Temp bins	With gradient
Log distance	-1.87*** (0.013)	-1.87*** (0.013)	-1.87*** (0.013)
Temp 25–30°C		0.011*** (0.002)	0.012*** (0.002)
Temp 30–35°C		0.008** (0.004)	0.009*** (0.004)
Temp 35–40°C		-0.020*** (0.006)	-0.018*** (0.006)
Temp 40–45°C		-0.053*** (0.007)	-0.051*** (0.007)
Temp 45–50°C		-0.066*** (0.009)	-0.062*** (0.010)
Log dist × temp gradient			-0.001* (0.0007)
Origin FE	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Observations	31,148,196	31,148,196	31,148,196
Pseudo R^2	0.87382	0.87386	0.87386

Notes: Standard errors clustered by origin and destination in parentheses. The small gradient term suggests limited systematic re-routing toward cooler areas during heat extremes.

G Illustrative climate-projection exercise (no-adaptation)

This section provides an illustrative projection under changing heat exposure. The exercise is intentionally conservative in interpretation: it assumes temperature changes but behavior and infrastructure do not (a strong assumption). We include it to translate estimated nonlinear response functions into an order-of-magnitude sense of how mobility could change if heat extremes become more frequent.

We use CMIP6 climate projections [Thrasher et al., 2022] to compute UTCI for 2073 under a mid-range warming scenario and then use the GAM to predict mobility under 2023 versus 2073 thermal conditions, holding other covariates fixed. The Lucas critique [Lucas Jr, 1976] is a reminder that behavioral rules can change with policy and technology; nevertheless, because daily travel often requires exposure to ambient conditions, mobility may remain a margin of adjustment even as mortality impacts adapt [Barreca et al., 2016, Heutel et al., 2021].

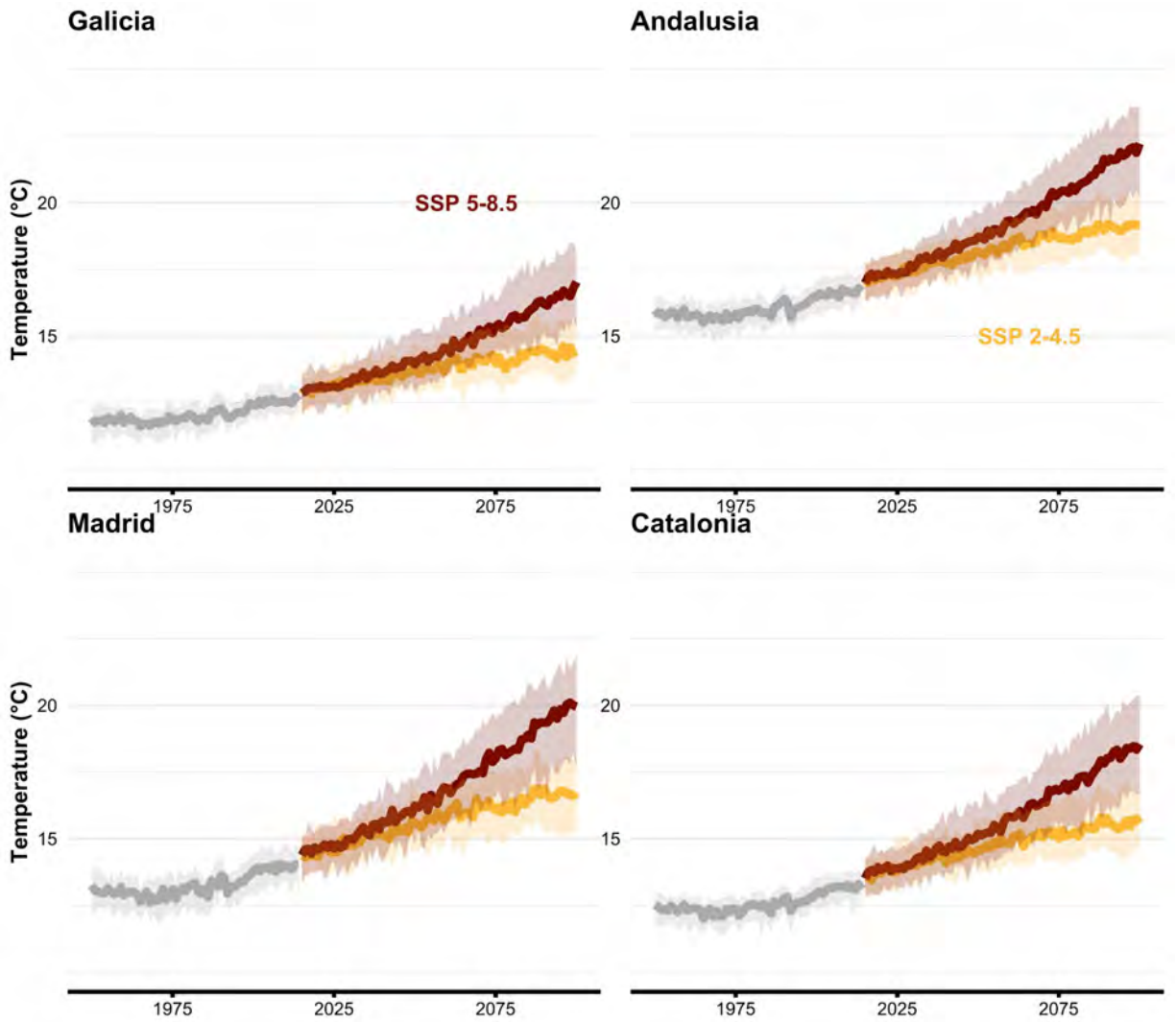


Figure A6: **CMIP6 projections for Spain (illustrative).** Southern regions (e.g., Andalusia) experience larger increases than northern regions; major metros (Barcelona, Madrid) also face substantial increases.

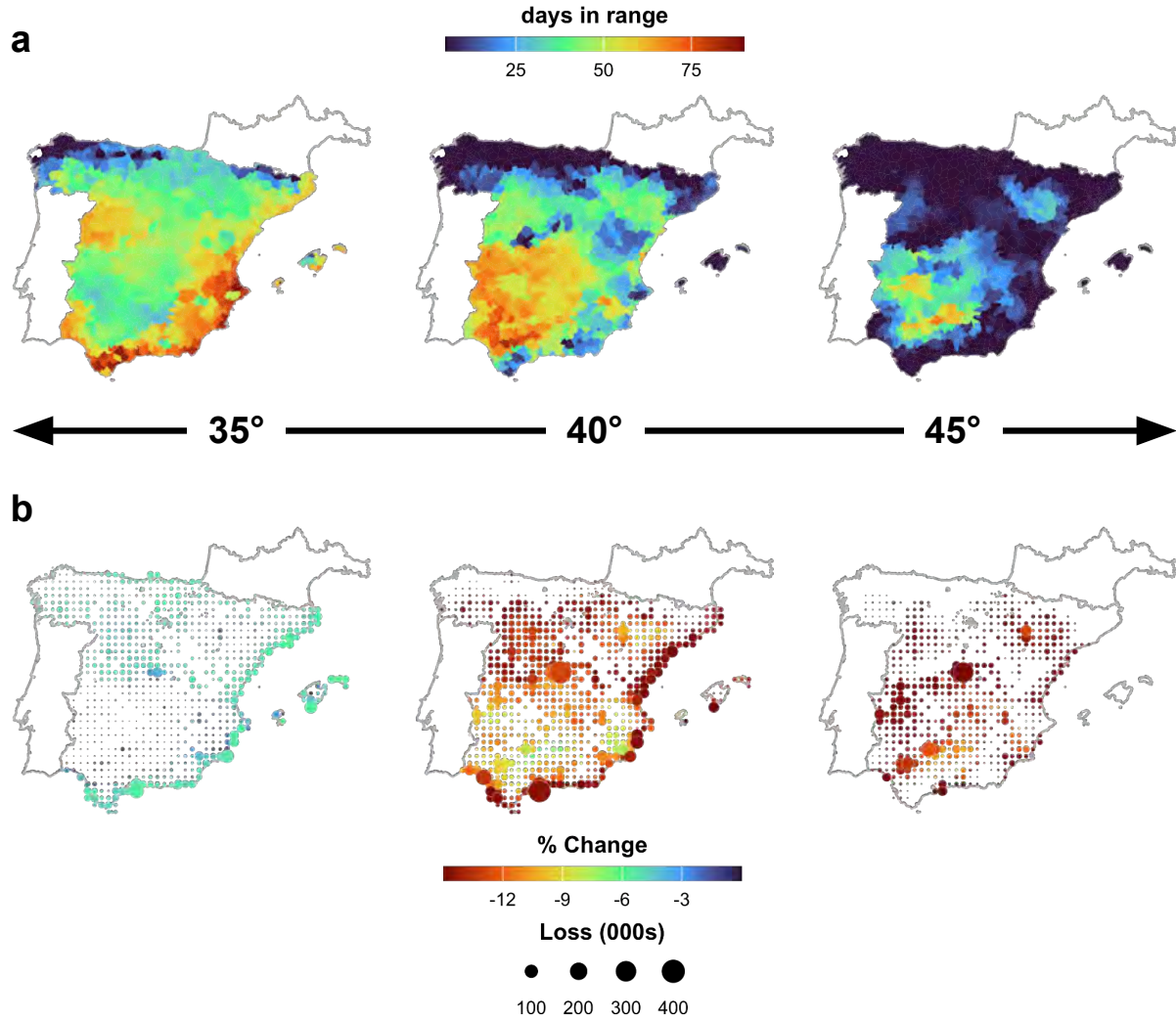


Figure A7: **Projecting mobility under future heat exposure (illustrative, no adaptation).** **A** Distribution shift: many southern districts experience far more days above 40 °C UTCL. **B** Predicted changes in mobility (relative and absolute) based on estimated nonlinear response functions.

Table A6: Gravity model interactions: destination population classes (average effects by destination group)

Temperature bin	Destination population class				
	Very low	Low	Medium	High	Very high
25–30° C	0.0267*** (0.0031)	0.0324*** (0.0045)	0.0321*** (0.0086)	0.0376*** (0.0091)	0.0482*** (0.0092)
30–35° C	0.0589*** (0.0049)	0.0629*** (0.0059)	0.0371*** (0.0078)	0.0277** (0.0080)	0.0224** (0.0084)
35–40° C	0.0862*** (0.0075)	0.0834*** (0.0095)	0.0350*** (0.0077)	-0.0007 (0.0076)	-0.0363*** (0.0077)
40–45° C	0.0791*** (0.0081)	0.0763*** (0.0108)	0.0086 (0.0083)	-0.0303** (0.0089)	-0.0874*** (0.0087)
45–50° C	0.0751*** (0.0158)	0.0588*** (0.0125)	0.0123 (0.0130)	-0.0783*** (0.0173)	-0.0962*** (0.0146)
$\log(d_{ij})$			-1.865***	(0.0112)	
Origin FE			Yes		
Destination FE			Yes		
Date FE			Yes		
Observations			19,638,009		

Notes: The most populous destinations experience the largest negative effects at high temperature bins, consistent with core activity losses.